PART 1 – report on building a mobile Network

You are a Network Engineer and you have been asked to set up a Wireless Network including Mobile devices (1) Laptop and (2) Phone.

You will simulate this network using Cisco Packet Tracer software.

When you have completed this report then you need to save this file in a folder with your Packet Tracer File. Zip this folder and upload to Moodle.

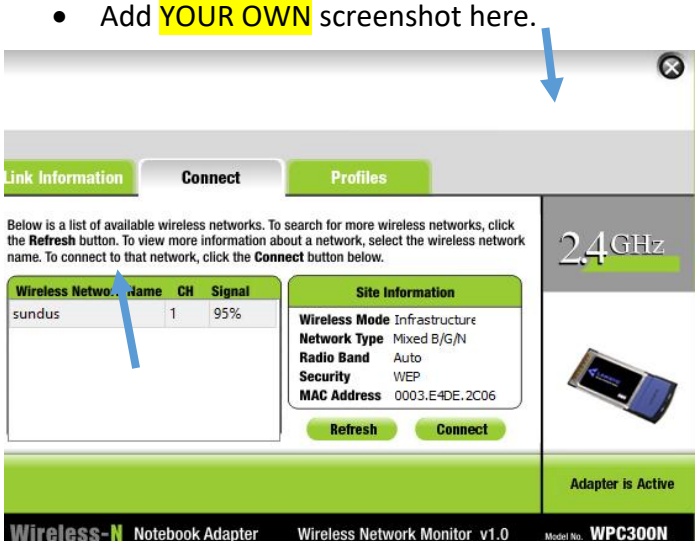
Step 1

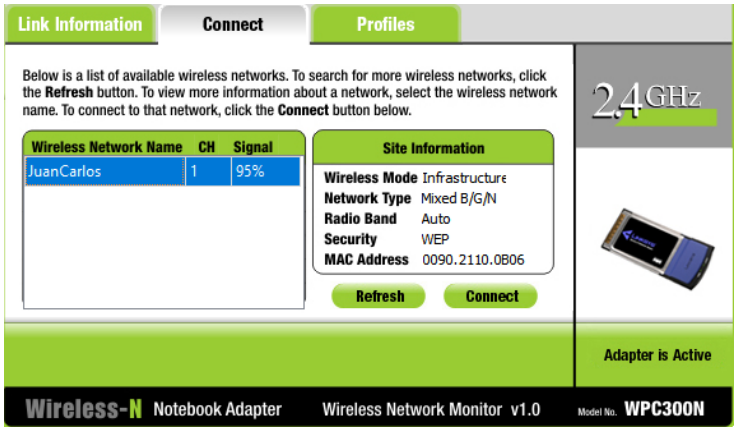
Create a Local Area Network using Packet Tracer Software

- In the left-hand corner select 'End Devices' and choose 1PC, 1 Phone & 1 Laptop,
- Choose a wireless router (WRT300N) and switch(2960-24TT)

Step 2

- Go to the laptop-> physical-> turn the laptop off and change the module to Linksys-WPC300N, turn the laptop back on.
- Go to the wireless router-> config, Wireless, tick WEP and add a WEP key (use 10 characters, hexadecimal only) eg……(BA123456CD)
- Go to GUI then set up and enable the DHCP server.
  The IP addresses will start from 100 and there will be a maximum of 50 IPs.
- Go to wireless and change the Network Name to your Name and save settings.
- Go back to the laptop -> desktop – PC->wireless and connect to your network using the WEP key you created.
- Add YOUR OWN screenshot here.





Step 3

Connecting your cabled devices

- Connect your wireless router to your switch using a copper-straight-through cable.
  - Router Ethernet 0/1 ……. switch FastEthernet0/1

- Connect your switch to your PC using a copper-straight-through cable.
  - Pc FastEthernet()…….switch FastEthernet 0/2

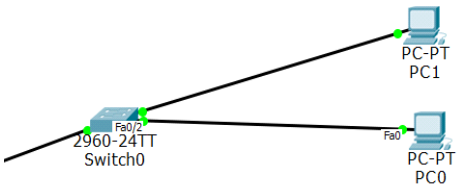
- Go to the PC->Desktop->IP configuration and enable DHCP in THE PC

Step 4

Mac address filtering – PICT OF SWITCH TO ONE PC ONLY

What is mac address filtering?

MAC address filtering is a network security method that controls device access by allowing or blocking connections based on their unique Media Access Control (MAC) address, essentially creating a digital whitelist (approved devices) or blacklist (blocked devices) on a router or firewall to manage who can join the network, adding a layer of access control beyond just passwords.



- **Go to your switch-> CLI add the commands below.**

Switch>ena

Switch#config t

Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport por
Switch(config-if)#switch port-security
Switch(config-if)#switch port-security max 1
Switch(config-if)#switch port-security mac-address sticky
Switch(config-if)#end

Next type the command

Switch#show port-security interface fa0/2

ENA

If there is no mac address ping the PC from the laptop. Then enter the command again.

Switch#show port-security interface fa0/2.
Record the Mac address.

000D.BDDD.748A:1

- Disconnect the PC from port fa0/2 and try to connect a new laptop to the same port fa0/2.
- Go to the new laptop – Desktop and Try to DHCP for a new IP address.

Step 5

Adding a mobile network

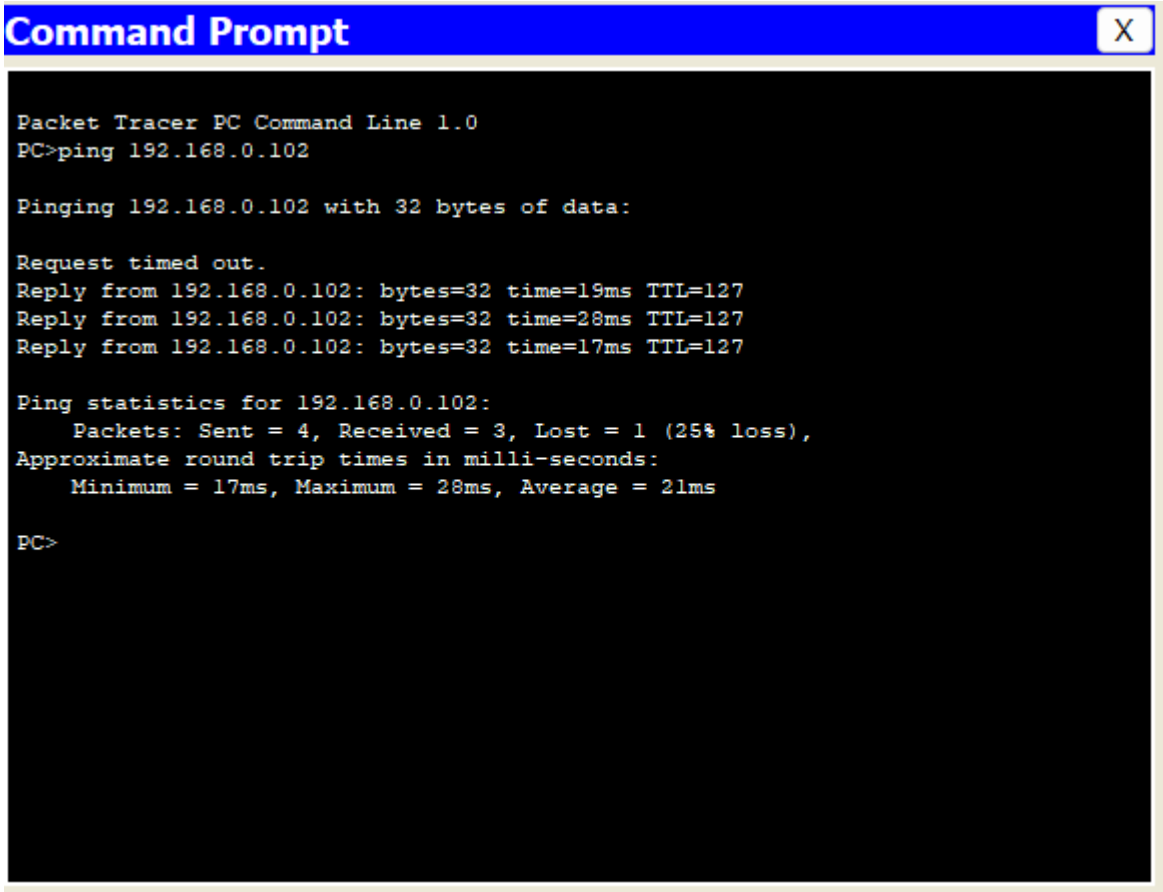
- From wireless device (ROUTER) choose a cell tower and a Central Services office. Connect these using a Coaxial cable.
    - Coaxial 0/0……coaxial 0/0
- Choose from end devices a smart phone.
- Connect the cell tower to the COS server with a coaxial cable and connect the COS to the router using a straight through cable internet…
    - backbone…Ethernet 0/2
- Go to your smart phone -> go to config and change the Display name to YourNamePhone

Step 6

Test your network.

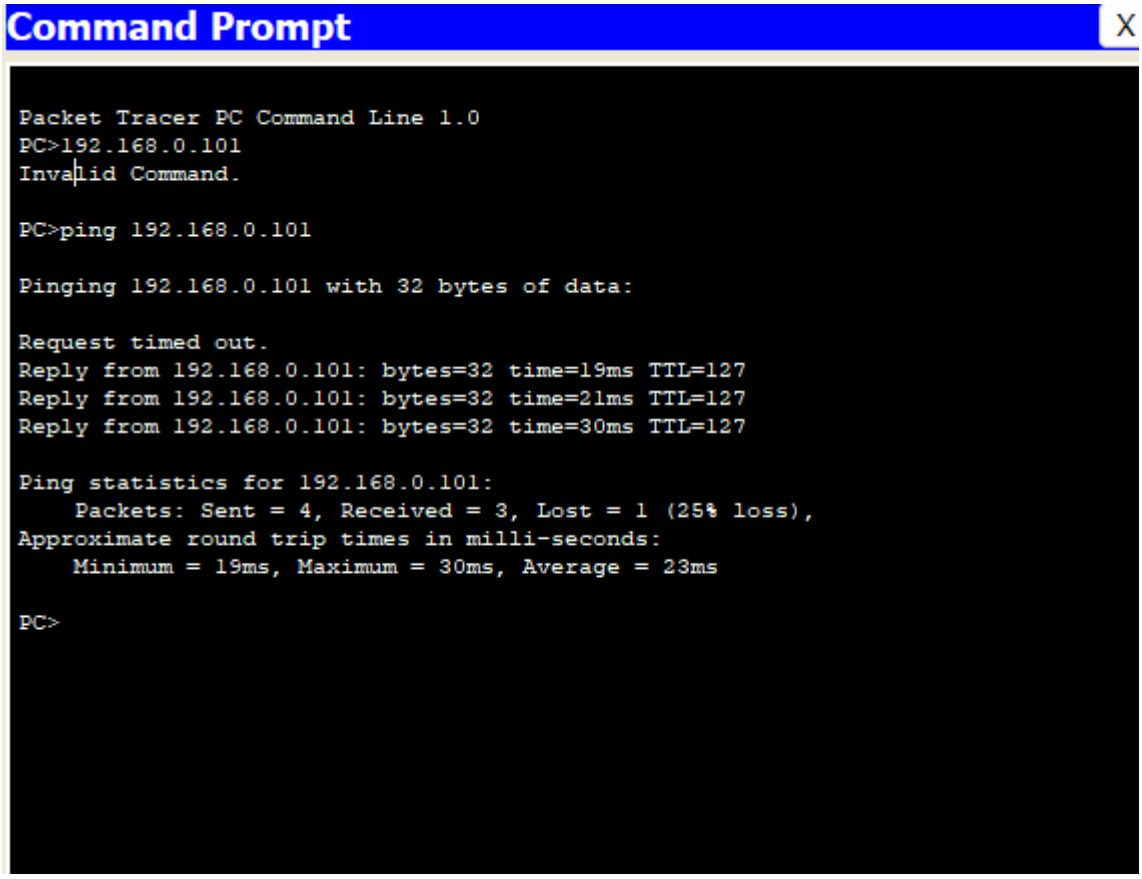
- Go to desktop-> command prompt.
- Ping the pc from your smartphone using the PC IP address.

Add Screen shot of reply below to replace the two here.



Ping the laptop from your smartphone using the IP address.

Record reply



Did all your packets transfer? Y/N

no

If not, what do you think may have happened?

I sent 4 packets received 3 and lost 1 .

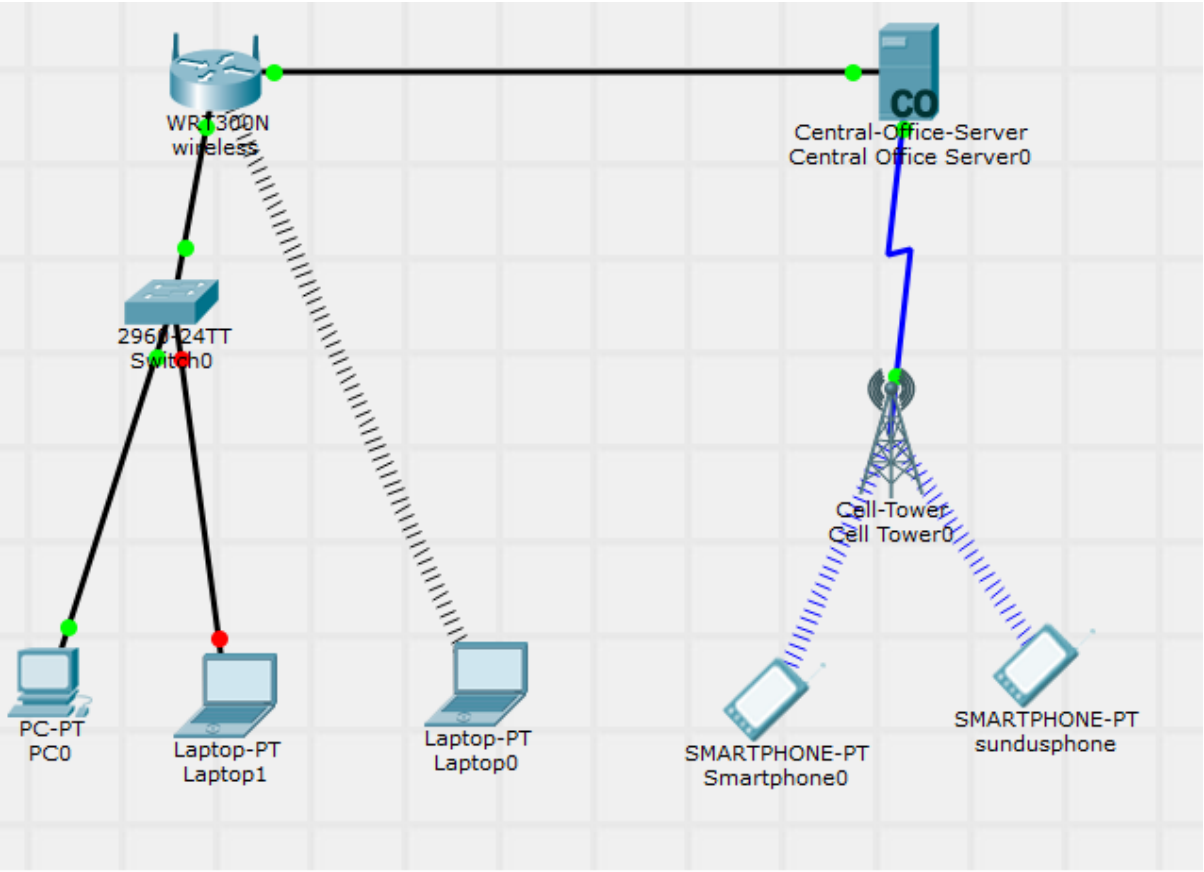
I think the network isnt strong enough to send all the packets.

Try again, is the result different, if so, what is different?

Yes the result is different i received back all the 4 packets i sent ,

I think the network was strong enough to send and receive them all.

****TAKE A SCREENSHOT OF YOUR COMPLETED NETWORK AND ADD BELOW**



PART 2a

What is DHCP? How is it used in your network? – use this information but answer in your own words.

Dynamic Host Configuration Protocol is an automatic network service that allows you to give your devices an automatic ip address.

I used it for my laptop devices and it generated an automatic ip address where if i wasn't using it i would have to assign an ip address each time.

What is a WEP key?

Wep key is a wired equivalent privacy key its a security password thats used to protect your wifi from unnecessary access.

I used the WEP KEY and i went to the wireless router-> config, Wireless, tick WEP and add a WEP key (use 10 characters, hexadecimal only) eg……(BA123456CD)

Wep key is always supposed to be hexadecimals and use letters and numbers.